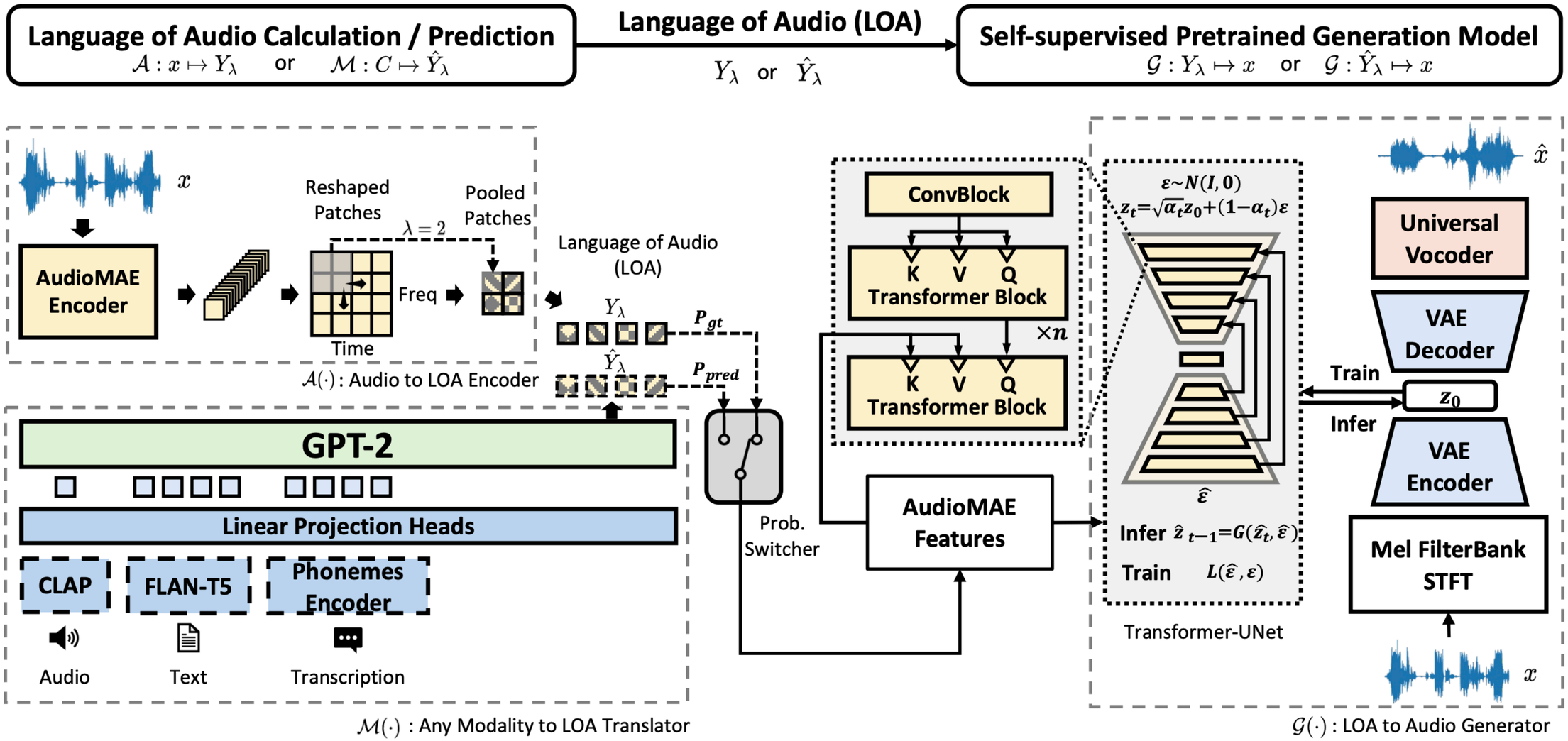
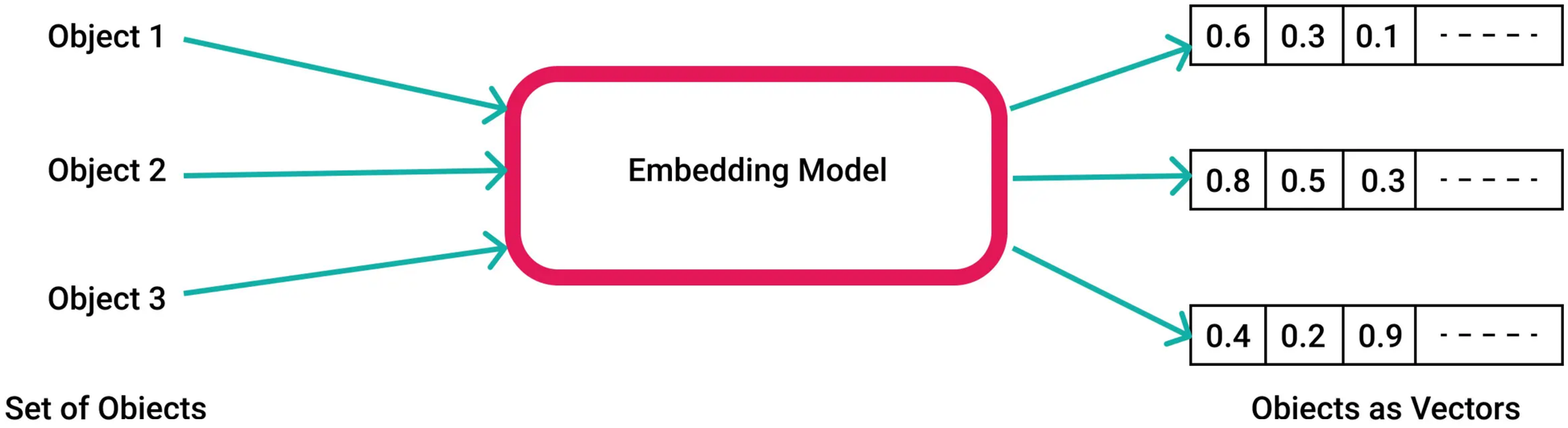
Audio LDM2

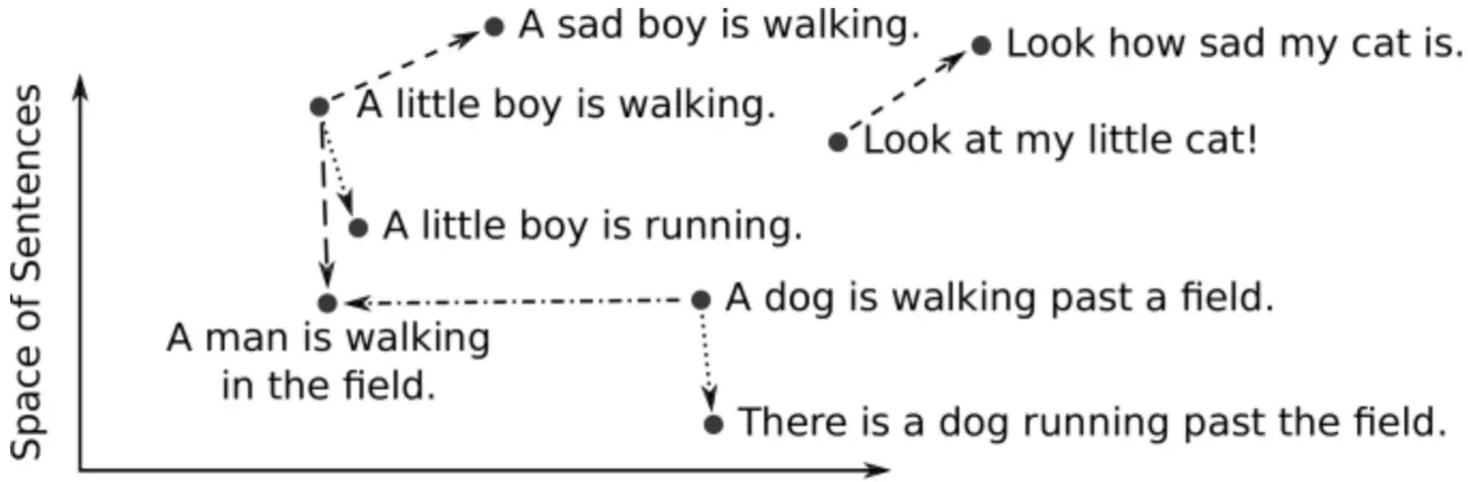
Audio LDM 2 - Architecture Overview



https://arxiv.org/pdf/2308.05734

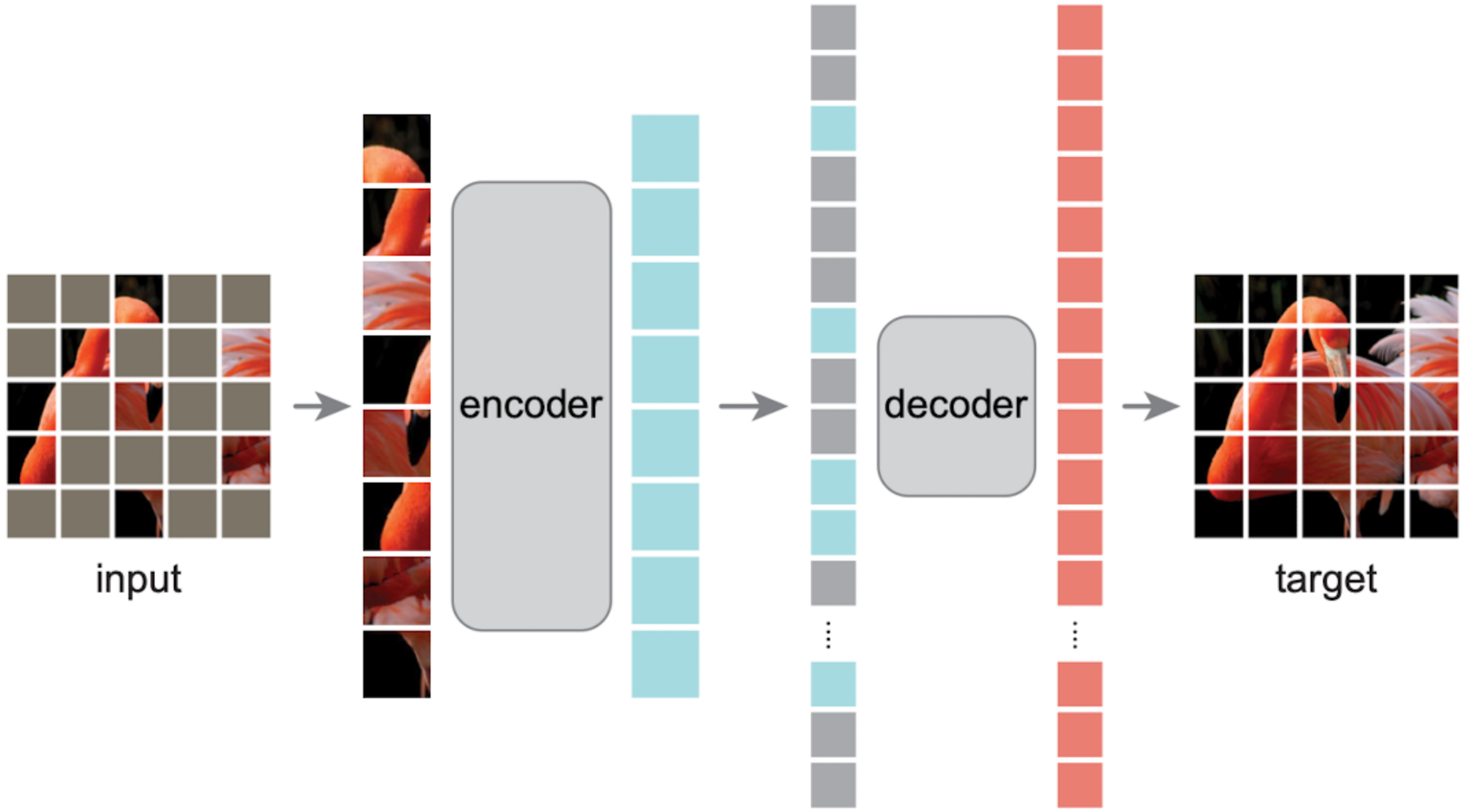
Embeddings





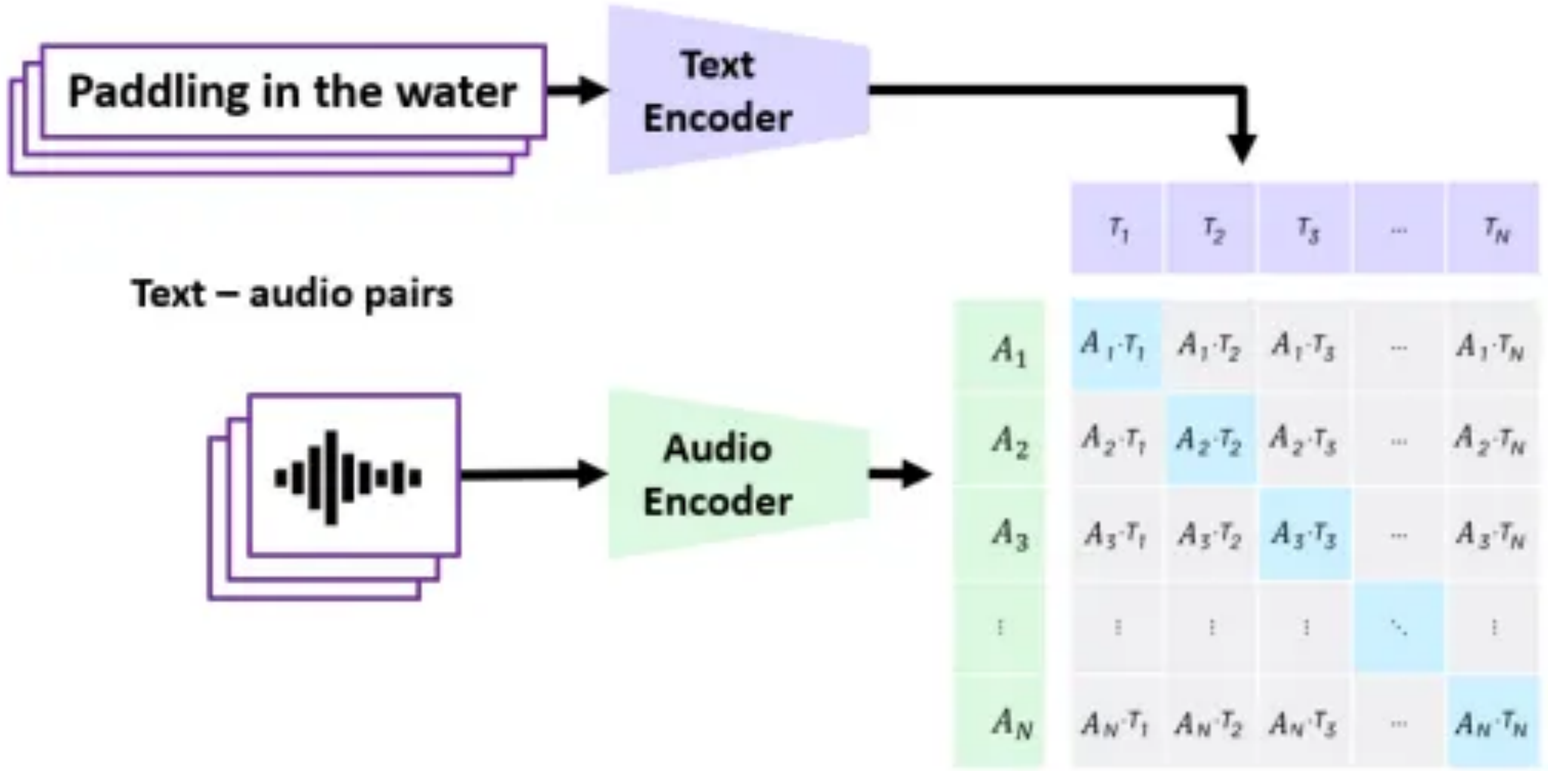
https://www.pinecone.io/learn/vector-embeddings/

Masked Autoencoder



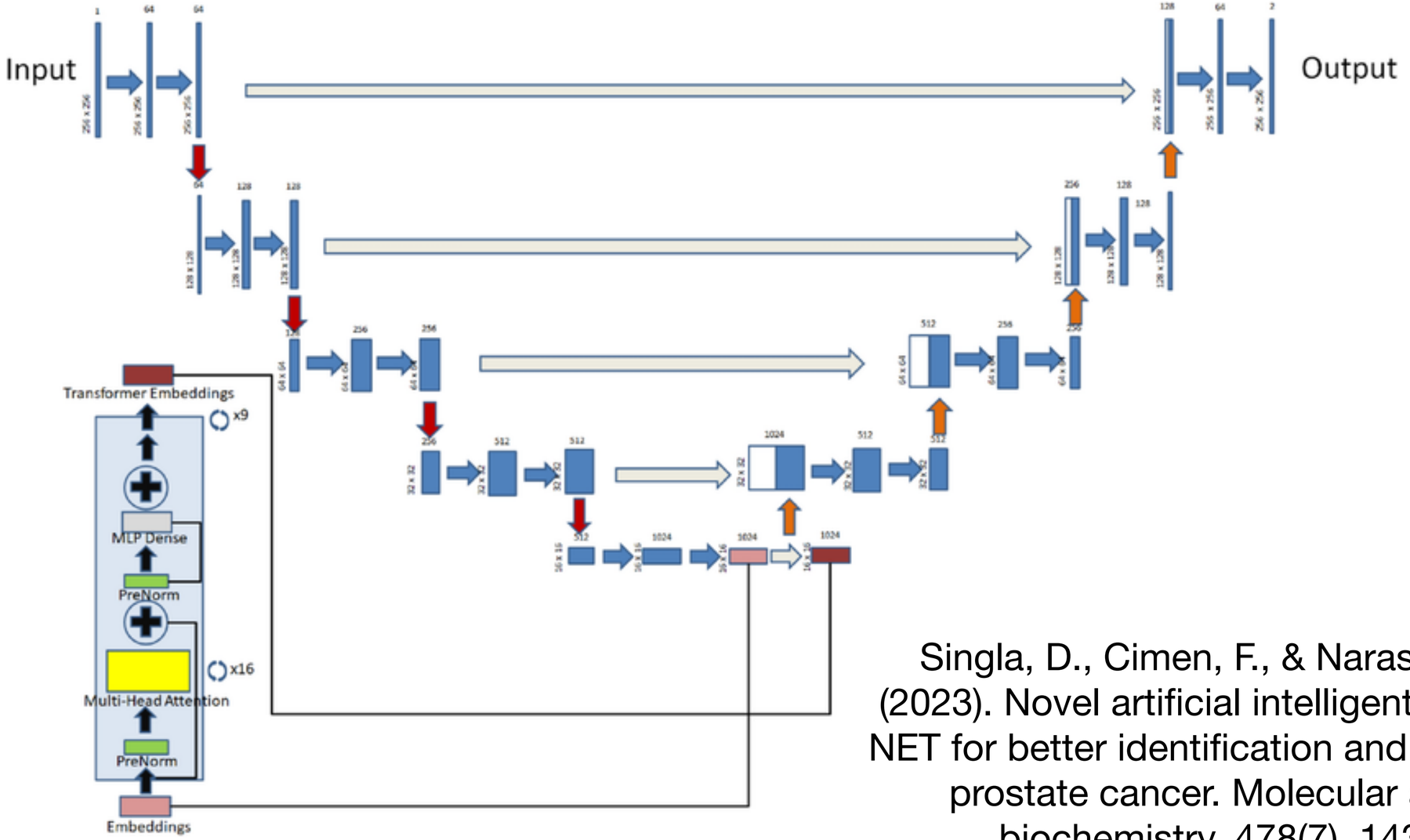
https://towardsdatascience.com/paper-explained-masked-autoencoders-are-scalable-vision-learners-9dea5c5c91f0/

CLAP - Contrastive Language Audio Pretraining



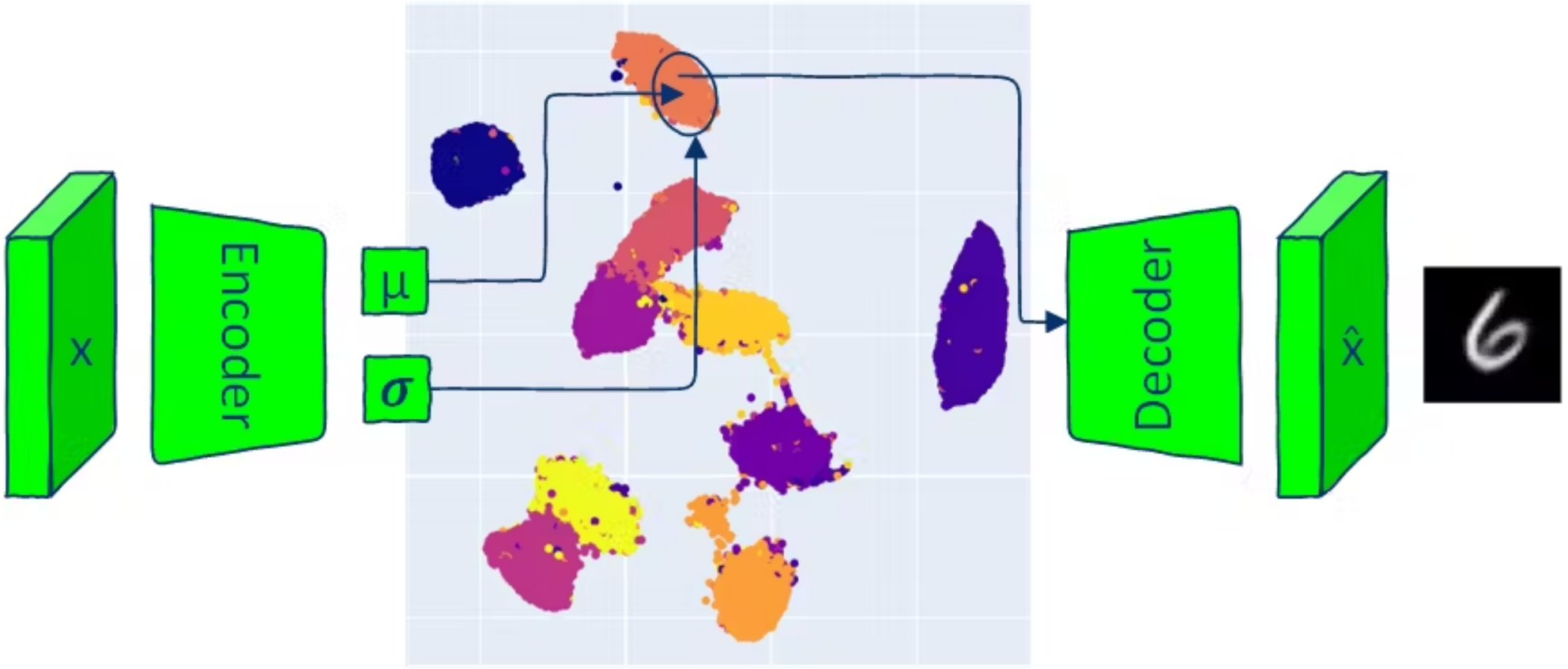
https://blog.gopenai.com/paper-in-short-clap-77be6efa9029

Transformer UNET



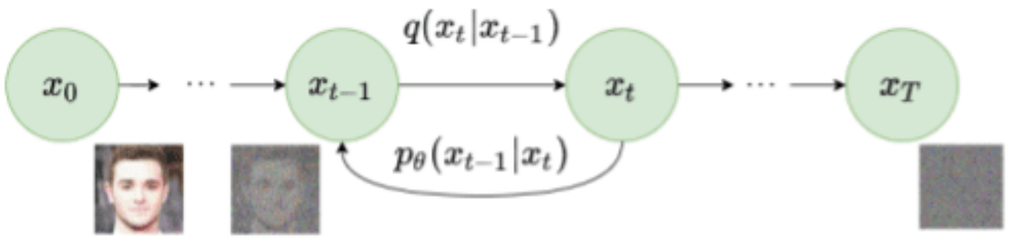
Singla, D., Cimen, F., & Narasimhulu, C. A. (2023). Novel artificial intelligent transformer U-NET for better identification and management of prostate cancer. Molecular and cellular biochemistry, 478(7), 1439-1445.

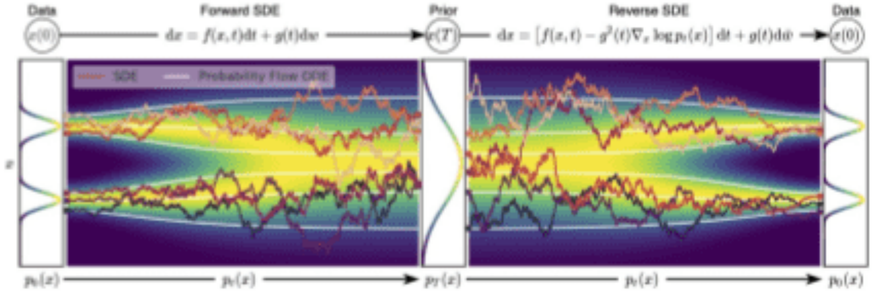
Variational Autoencoder

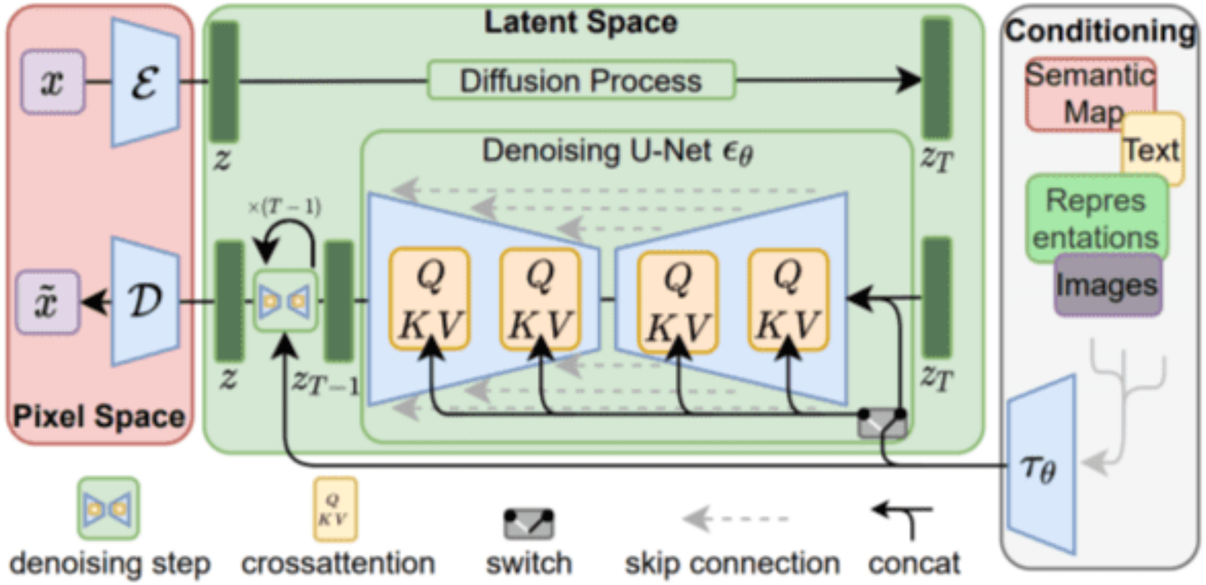


From https://hackernoon.com/how-to-sample-from-latent-space-with-variational-autoencoder

Latent Diffusion Model







Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
6: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

https://theaisummer.com/diffusion-models/